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M1 – Method mark, awarded for correct mathematical technique, independent of numbers used.

A1 – Answer mark, awarded for a correct result to a calculation, dependent on M

B1 – Independent mark, can be awarded regardless of other marks, for a correct method

E1 – Explanation mark, for correct use of terminology and mathematics for justification

F – Follow through from an incorrect result

AWRT – Answers which round to a given number

AWFW – Answers which fall within a given range

ACF – Any correct form, answers which are mathematically equivalent to the given answer

OE – Or Equivalent, numbers expressed in different forms

NE – not enough, not awardable/insufficient

A candidate may only lose a maximum of one mark throughout the paper for omission of units

Q	Marking Instructions	Marks	Notes
1 (i)	$\frac{2x+2}{4x^2-4} - \frac{2x-2}{4x^2-4}$ $= \frac{4}{4x^2-4}$ $= \frac{1}{x^2-1}$	**M1** **A1**	**For both fractions correct with the same denominator** **Clearly shown result** **A student may use partial fraction decomposition, which is worthy of both marks**
	Sub Total	**2**	
1 (ii)	(A) $2\sum_{r=2}^{n} \frac{1}{2x-2} - \frac{1}{2x+2}$ $r = 2$ $\mathbf{\frac{1}{2}} - \frac{1}{6}$ $r = 3$ $\mathbf{\frac{1}{4}} - \frac{1}{8}$ $r = 4$ $\frac{1}{6} - \frac{1}{10}$ Bold will remain $r = n-1$ $\frac{1}{2n-4} - \mathbf{\frac{1}{2n}}$ $r = n$ $\frac{1}{2n-2} - \mathbf{\frac{1}{2n+2}}$ Resulting formula $\frac{1}{2} + \frac{1}{4} - \frac{1}{2n} - \frac{1}{2n+2}$ $\frac{3}{2} - \left(\frac{1}{n} + \frac{1}{n+1}\right)$ (B) $\frac{3}{2}$	**M1** **M1** **M1** **A1** **A1**	**Clearly rewriting the sum, with a 2 factored in the front** **An attempt to use the method of differences with at least one term cancelled** **For a halved formula (ACF)**
	Sub Total	**5**	
	Question Total	**7**	

Q	Marking Instructions	Marks	Notes
2 (i)	For these planes to be perpendicular, their normal vectors must be perpendicular. Therefore: $4a - 10 - 6 = 0$ $a = 4$	**E1** **M1** **A1**	**An equation in terms of a**
	Sub Total	**3**	
2 (ii)	In matrix form, $\begin{pmatrix} 4 & 5 & -2 \\ 4 & -2 & 3 \\ 7 & 1 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 48 \\ 74 \\ 43 \end{pmatrix}$ $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -7 & -12 & 11 \\ 13 & 22 & -20 \\ 18 & 31 & -28 \end{pmatrix}\begin{pmatrix} 48 \\ 74 \\ 43 \end{pmatrix}$ $(-751, 1392, 1954)$	**M1** **M1** **A1**	**f.t their a** **Pre-multiplying by the inverse matrix** **Accept column vector form**
	Sub Total	**3**	
	Question Total	**6**	

Q	Marking Instructions	Marks	Notes
3 (i)	(A) $D = \begin{pmatrix} 1 & 4 \\ 6 & 2 \end{pmatrix}\begin{pmatrix} 3 & 1 \\ a & 5 \end{pmatrix}$ $D = \begin{pmatrix} 4a+3 & 21 \\ 2a+18 & 16 \end{pmatrix}$ (B) Area of triangle = 28 units $\det D = 64a + 48 - 42a - 378$ $= 22a - 330$ Area of triangle after = $616a - 9240$	**M1** **A1** **B1** **B1** **A1**	**Setting up the equation in the correct order**
	Sub Total	**5**	
3 (ii)	For an invariant line, points remain on the line, so $\begin{pmatrix} 1 & 4 \\ 6 & 2 \end{pmatrix}\begin{pmatrix} x \\ mx \end{pmatrix} = \begin{pmatrix} x' \\ mx' \end{pmatrix}$ $x + 4mx = x'$ $6x + 2mx = mx'$ $6x + 2mx = mx + 4m^2x$ LHS = RHS, therefore $6 + 2m = m + 4m^2$ $4m^2 - m - 6 = 0$ $y = \frac{1-\sqrt{97}}{8}x$ $y = \frac{1+\sqrt{97}}{8}x$	**B1** **M1 dep** **M1** **A1**	**Setting up an equation including matrices** **Forming simultaneous equations** **Comparing coefficients** **Do not accept m=…, the answers must be lines.** **f.t their equation for m** **Accept methods which use eigenvalues and eigenvectors**
	Sub Total	**4**	
	Question Total	**9**	

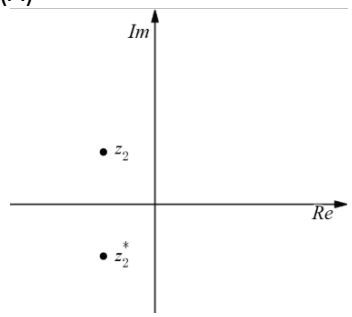
Q	Marking Instructions	Marks	Notes
4	Converting to cartesian form $(x-4)^2+(y+2)^2=25$	**B1**	
	$\tan\left(\frac{\pi}{4}\right)=1$ $y=x+c$ $-1=0+c$ $y=x-1$	**B1**	
	$(x-4)^2+(x+1)^2=25$ $x^2-8x+16+x^2+2x+1=25$ $2x^2-6x-8=0$ $x=4,-1$ $y=3,-2$ $4+3i,-1-2i$	**M1** **A1**	**Attempting to solve** **Potential complex numbers stated**
	$\arg z+i$ is a half line from the line y=0, so $-1-2i$ is not feasible	**E1**	**Selecting correct solution**
	$Im(4+3i)=3$	**A1**	
	Question Total	**6**	

Q	Marking Instructions	Marks	Notes
5 (i)	For the cubic to have real coefficients, the complex solutions must be conjugates of each other, since it causes the imaginary part to cancel when expanded.	**E1**	**Reference to conjugate, and that imaginary parts cancel**
	Sub Total	**1**	
5 (ii)	$z = 2 \pm \sqrt{3}i$ $z - 2 = \sqrt{3}i$ $(z-2)^2 + 3 = 0$ $(z^2 - 4z + 7)$	**M1**	**Finding the quadratic factor**
	$(z^2 - 4z + 7)(z - k)$	**M1**	**Forming the quadratic and linear factor together, with an attempt to expand**
	$z^3 - (4 + k)z^2 + (7 + 4k)z - 7k$	**A1**	**ACF** **Accept a=..., b=…, c=…** **Accept answers that use Vieta's formulas**
	Sub Total	**3**	
	Question Total	**4**	

Q	Marking Instructions	Marks	Notes
6	First, I will test for $n = 1$, my base case $$\sum_{r=1}^{1} 3r^2 = 3(1)^2 = 3$$ $$\frac{1}{2}1(2)(3) = 3$$ The statement is true for $n = 1$	**B1**	
	I will now assume the statement is true for some $n = k$, where k is a natural number Therefore, $$\sum_{r=1}^{k} 3r^2 = \frac{1}{2}k(k+1)(2k+1)$$ My inductive hypothesis	**M1**	**Assumption made**
	Test for $n = k + 1$ $$\sum_{r=1}^{k+1} 3r^2 = 3(1)^2 + 3(2)^2 + \cdots + 3(k)^2 + 3(k+1)^2$$ $$= \sum_{r=1}^{k} 3r^2 + 3(k+1)^2$$ Via my inductive hypothesis $$\frac{1}{2}k(k+1)(2k+1) + 3(k+1)^2$$ $$\frac{1}{2}(k+1)(2k^2 + k + 6k + 6)$$ $$\frac{1}{2}(k+1)(2k^2 + 7k + 6)$$ $$\frac{1}{2}(k+1)(k+2)(2k+3)$$	**M1** **A1**	
	Therefore, it is true for $n = k + 1$, given it is true for $n = k$. I have also shown it to be true for $n = 1$. Therefore, via the principal of mathematical induction, I have proved this statement true for all $n \in \mathbb{N}$ (Q.E.D)	**A1**	**Clear justification**
	Question Total	**5**	

Q	Marking Instructions	Marks	Notes
7 (i)	$30°$ clockwise = $330°$ anti-clockwise $\begin{pmatrix}\cos 330 & -\sin 330\\ \sin 330 & \cos 330\end{pmatrix}$ $\boldsymbol{M} = \begin{pmatrix}\frac{\sqrt{3}}{2} & \frac{1}{2}\\ -\frac{1}{2} & \frac{\sqrt{3}}{2}\end{pmatrix}$ $\det \boldsymbol{M} = \mathbf{1}$	**M1** **A1**	**Accept use of the clockwise rotation matrix** **For both M and det M**
	Sub Total	**2**	
7 (ii)	$\boldsymbol{M^{-1}}$ is a rotation $30°$anticlockwise $\det \boldsymbol{M^{-1}} = \frac{1}{\det \boldsymbol{M}} = 1$ $\boldsymbol{M^{-1}}$ **preserves area and orientation of vertices**	**A1** **M1** **A1**	
	Sub Total	**3**	
	Question Total	**5**	

Q	Marking Instructions	Marks	Notes
8	$\cos\theta = \dfrac{u \cdot v}{\lvert u\rvert \times \lvert v\rvert}$		
	$\lvert u\rvert = \sqrt{\alpha^2 + \beta^2 + \gamma^2}$	**B1**	
	$\lvert v\rvert = 5\sqrt{3}$	**B1**	
	$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$	**M1**	
	$= (4)^2 - 2(-5) = 26$ $\lvert u\rvert = \sqrt{26}$	**M1**	
	$u \cdot v = 5\alpha + 5\beta + 5\gamma = 5(\alpha + \beta + \gamma)$ $= 5(4) = 20$	**M1**	
	$\dfrac{20}{\sqrt{26} \times 5\sqrt{3}} = \cos\theta$	**M1**	
	$\theta \approx 63.07°$	**A1**	
	Question Total	**7**	

Q	Marking Instructions	Marks	Notes
9 (i)	(A) $\frac{1}{2-7i} = \frac{2+7i}{(2-7i)(2+7i)}$	**M1**	**Calculations must be clear**
	$\frac{2}{53} + \frac{7}{53}i$	**A1**	**Expressed as a complex number**
	(B) $\left\|\frac{1}{z}\right\| = \frac{\sqrt{53}}{53}$	**B1**	**ACF**
	$\arg\frac{1}{z} = \arctan\left(\frac{7}{2}\right) \approx 1.2925\ rad$	**B1**	
	$\frac{\sqrt{53}}{53}(\cos(1.2925) + i\sin(1.2925))$	**A1**	**ACF**
	Sub Total	**5**	
9 (ii)	(A)  	**A1**	**Accept without labelling** **Angle must be approximately $45°$ by eye (or labelled in radians)** **The points must lie on the same vertical line (or very close to) and be equidistant from the real axis (or very close to**
	(B) $5\left(\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right)$	**A1**	
	Sub Total	**2**	
9 (iii)	(A) $\left\|\frac{z_2^*}{z_1}\right\| = \left\|z_2^* \times \frac{1}{z_1}\right\|$	**M1**	
	$\frac{5\sqrt{53}}{53}$	**A1**	**ACF**
	(B) $\arg(z_2 \times z_1) = \arg\left(z_2 \div \frac{1}{z_1}\right)$	**M1**	**Accept finding arg of z_1**
	$\frac{3\pi}{4} - 1.2925 \approx 1.06$	**A1**	**AWFW 1.055 – 1.068**
	Sub Total	**4**	
	Question Total	**11**	